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PROTECTING INTELLECTUAL PROPERTY USING DIGITAL SIGNATURES

Abstract

Methods of protecting intellectual property using digital signatures include generating reference digital signatures of first content, generating a digital signature of second content, comparing the digital signature of the second content to the reference digital signatures to identify matching reference digital signatures, and selectively performing an action based on the matching reference digital signatures and a policy. The first content may include proprietary information of an organization and/or posts of a machine-learning (ML) model. The second content may include source code intercepted from a transmission directed to an external site, such as a ML model, and/or source code saved to a source code repository. Actions may include, without limitation, initiating an audit of the second content, sending a notification to a user interface indicating that the second content likely contains a portion of the first content, releasing the transmission, and/or terminating the transmission.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure generally relates to a computer security system. In particular, the present disclosure relates to protecting intellectual property through the use of digital signatures.

BACKGROUND

[0002] Generative artificial intelligence machine learning (AIML) models, such as large language models (LLMs), are rapidly proliferating. LLMs are language models built upon neural networks containing billions of parameters trained to generate textual content, including computer source code.

[0003] Publicly-accessible LLMs pose privacy and security risks. For example, an employee of an organization may submit proprietary source code to a publicly-accessible LLM for training purposes and/or to improve the source code. An LLM may, however, provide an indication of and/or disclose information on which the LLM was trained.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] The disclosure will be understood more fully from the detailed description given below and from the accompanying figures of embodiments of the disclosure. The figures are used to provide knowledge and understanding of embodiments of the disclosure and do not limit the scope of the disclosure to these specific embodiments. Furthermore, the figures are not necessarily drawn to scale.

[0005] FIG. 1 is a block diagram of a computing platform for protecting intellectual property through the use of digital signatures, according to an embodiment.

[0006] FIG. 2 illustrates a method of protecting intellectual property through the use of digital signatures, according to an embodiment.

[0007] FIG. 3 is another block diagram of the computing platform, according to an embodiment.

[0008] FIG. 4 illustrates another method of protecting intellectual property through the use of digital signatures, according to an embodiment.

[0009] FIG. 5 is another block diagram of the computing platform, according to an embodiment.

[0010] FIG. 6 illustrates another method of protecting intellectual property through the use of digital signatures, according to an embodiment.

[0011] FIG. 7 is another block diagram of the computing platform, according to an embodiment.

[0012] FIG. 8 depicts a diagram of an example computer system in which embodiments of the present disclosure may operate.

DETAILED DESCRIPTION

[0013] Aspects of the present disclosure relate to protecting intellectual property through the use of digital signatures.

[0014] An organization/entity may generate proprietary information (i.e., intellectual property) related to products, services, and/or customers, which may include designs/inventions (e.g., source code and/or circuit designs), marketing materials, customer lists, employee information, legal documents, and/or other content, and/which may be embodied as digitized text, images, video, and/or audio.

[0015] An organization may want to identify proprietary information contained in outgoing

correspondence and/or identify internal contributions to proprietary information, such as to identify contributions of a machine-learning (ML) model (e.g., ML-generated source code). An organization may also want to perform automated actions (e.g., initiate audits, issue warnings, and/or block transmissions of content to external networks). The sheer quantity and complexity of proprietary information may make it technically challenging to determine whether such content includes proprietary information, and/or to identify internal contributions (e.g., of a ML model).

[0016] In an example, protecting intellectual property through the use of digital signatures includes generating a database of digital signatures of proprietary content, generating a digital signature of a user post directed to an external network, comparing the digital signature of the user post to the database of digital signatures to determine if the user post contains or likely contains proprietary information, and selectively performing one or more actions configurable policies if the user post contains or likely contains proprietary information. The digital signatures may be generated using, for example and without limitation, piecewise hashing, rolling hashes, Merkle trees, Simhash, Minhash, content defined chunking (CDC), differential hashing, shingling, and/or combinations thereof. The one or more actions may be performed by a processing device and include, without limitation, initiating an audit of the user post, terminating transmission of the user post, alerting a user that the user post contains or likely contains proprietary content, and/or permitting the user to terminate transmission of the user post.

[0017] In another example, protecting intellectual property through the use of digital signatures includes generating a database of digital signatures of posts/outputs of a ML model (e.g., an internal/proprietary ML model and/or an external/public ML model), generating a digital signature of source code submitted to a source code repository and comparing it to the database of digital signatures to quickly and efficiently identify source code that contains or likely contains posts of the ML model, and selectively performing one or more actions based on the comparison. Actions may include, without limitation, initiating an audit of the source code and/or alerting a user that the source code contains or likely contains posts of the ML model.

[0018] Protecting intellectual property through the use of digital signatures may further include configurable policies. In an example, configurable policies define one or more types of actions and criteria that specify conditions (e.g., thresholds) for performing the actions. In another example, configurable policies specify one or more types of digital signatures to generate and/or to compare, and/or one or more levels of granularity at which to generate and/or compare digital signatures. In another example, configurable policies specify content for which digital signatures are to be captured, a frequency (e.g., a schedule) at which a digital signature database is to be updated. In another example, configurable policies relate to IP (e.g., content) leakage risk identification.

[0019] Technical advantages of the present disclosure include, but are not limited to, improvements in functions/operations of a computing platform. For example, and without limitation, although digital signatures may be computed based on the entire contents of computer files. For relatively large files and/or relatively numerous files, however, this may require significant computational resources and storage capacity. As disclosed herein, digital signatures may be computed based on portions (i.e., snippets) of file contents, descriptors of files and/or file directory structures, and/or metadata, at one or more configurable/selectable levels of granularity (e.g., based on configurable policies), with fewer computational resources and reduced storage capacity. Levels of granularity may be configured/selected to control accuracy versus efficiency (i.e., higher accuracy may utilize more computing resources/time).

[0020] Technical advantages of the present disclosure further include, but are not limited to, improvements in functions/operations of automated tools for protecting intellectual property, including automated audit initiation, automated user notifications, and/or automated pausing/termination of user posts directed to external networks.

[0021] FIG. 1 is a block diagram of a computing platform **100** for protecting intellectual property through the use of digital signatures, according to an embodiment. Computing platform **100** may

include circuitry (e.g., logic gates and/or a processor and memory). Computing platform **100** may represent a single computing platform or multiple computing platforms, which may be centralized or distributed. An example of computing platform **100** is provided further below with reference to FIG. **8**.

[0022] In the example of FIG. **1**, computing platform **100** includes a signature generator **102** that generates digital signatures **104** of reference content **106**, and a digital signature(s) **108** of user content **110**. Reference content **106** and user content **110** may include, for example and without limitation, digitized human-readable content (e.g., source code), binary files (e.g., machine-readable code compiled from source code), digitized images, digitized video, and/or digitized audio.

[0023] Computing platform **100** further includes a signature matching engine **116** that compares digital signature **108** of user content **110** to digital signatures **104** of reference content **106**, and outputs match results **118**.

[0024] Computing platform **100** further includes an action tool **120** that selectively performs one or more actions based on match results **118**. As an example, and without limitation, action tool **120** may initiate an audit of user content **210** if match results **118** meet a criterion. Additional examples are provided further below.

[0025] Computing platform **100** may further include a signature database **112** that stores digital signatures **104**, a user interface (UI) **114** that receives user content **110** from a user, and/or a policy manager **122** that manages policies **124**, **126**, and **128** for signature generator **102**, signature matching engine **116**, and action tool **120**, respectively.

[0026] FIG. **2** illustrates a method **200** of protecting intellectual property through the use of digital signatures, according to an embodiment. Method **200** is described below with reference to FIG. **1**. Method **200** is not, however, limited to the example of FIG. **1**.

[0027] At **202**, signature generator **102** generates digital signatures **104** of reference content **106**.

[0028] Signature generator **102** may generate digital signatures **104** for units of reference content **106**, where a unit of reference content **106** may correspond to a single computer file or a group of related computer files (e.g., arranged in hierarchical and/or directory-based structures). As an example, and without limitation, source code may include multiple related/linked computer files (e.g., libraries), which may be treated as a unit of reference content **106**.

[0029] Signature generator **102** may generate digital signatures **104** at one or more levels of granularity. At a most detailed level of granularity, signature generator **102** may generate a digital signature **104** based on the entire contents of a unit of reference content **106**. At less granular levels of granularity, signature generator **102** may construct a string of information/features related to a unit of reference content **106**, and may generate a digital signature **104** of the string. Features may include, for example and without limitation, one or more portions (i.e., segments or snippets) of contents of a unit of reference content **106** and/or metadata/descriptors of the unit of reference content **106**. Metadata/descriptors may include, for example, and without limitation, file names, file size, time/date stamps, authors, descriptions of directory structures, and/or other information. Signature generator **102** may include a feature extractor **103** that extracts features of reference content **106** and arranges the extracted features in a string based on policies **124**.

[0030] Signature generator **102** may generate multiple digital signatures **104** for a unit of reference content **106**, at multiple respective levels of granularity. Signature generator **102** may, for example, construct multiple strings for a unit of reference content **106**, where a first string (i.e., a least granular string) includes relatively few features (e.g., relatively few and/or relatively short snippets, and/or relatively little/few metadata/descriptors), and where each subsequent string contains progressively more features (e.g., more and/or longer snippets and/or more metadata/descriptors), and/or various combinations of features. Signature generator **102** may generate a digital signature **104** for each string. Levels of granularity may be configurable (e.g., via policies **124** and **126**). Levels of granularity may be configured/selected to control accuracy versus efficiency (i.e., higher

accuracy/more granularity may utilize more computing resources/time).

[0031] Signature generator **102** may compute one or more types of digital signatures based on one or more encoding schemes, such as a cyclic redundancy check (CRC) code, and/or a hash function that maps data of arbitrary size to a fixed-size value, which may be referred to as a hash value, a hash code, a hash digest, a digest, or a hash. Signature generator **102** is not limited to CRC codes or hash functions.

[0032] Signature generator **102** may compute digital signatures based on policies **124**, which may specify an encoding scheme(s) and features to use in determining digital signatures.

[0033] At **204**, signature generator **102** generates digital signature(s) **108** of user content **110**, such as described above with respect to digital signatures **104**. In one embodiment, the digital signature generation method for user content **110** is the same as the digital signature generation method for reference content **106**.

[0034] At **206**, signature matching engine **116** compares digital signature **108** of user content **110** to digital signatures **104** of reference content **106** to identify digital signatures **104** that match digital signature(s) **108** (i.e., matching digital signatures).

[0035] Where signature generator **102** determines digital signatures **104** and **108** based on descriptors, metadata, and/or snippets, the digital signatures may not necessarily be unique. In such a situation, signature matching engine **116** may identify matching digital signatures for which the corresponding proprietary content **106** does not precisely match user post **110**. In this example, match results **118** may indicate a quantity/number of matching digital signatures, which may be indicative of a likelihood that user content **110** includes a portion of reference content **106**.

[0036] Where signature generator **102** determines multiple types of digital signatures and/or digital signatures at multiple levels of granularity, signature matching engine **116** may initially compare digital signatures of a first type or first level of granularity (e.g., a least granular level) (e.g., based on policies **126**). If the quantity/number of matching digital signatures meets a threshold, signature matching engine **116** may compare digital signatures **104** and **108** corresponding to a lower/more-detailed level of granularity to reduce the number/quantity of matching digital signatures.

[0037] Signature generator **102** and signature matching engine **115** may utilize features/methods of signature/snippet/codeprint matching tools designed to detect open-source code within source code of an organization, and/or signature matching tools designed to detect security vulnerabilities in source code.

[0038] At **208**, action tool **120** selectively performs one or more actions based on match results **118** and policies **128**, examples of which are provided further below.

[0039] FIG. **3** is another block diagram of computing platform **100**, according to an embodiment.

In the example of FIG. **3**, reference content **106** includes proprietary content **106A** of an organization, and user content **110** includes a user post **110A** directed to an external site **330** via an external network **332** (e.g., the Internet). Computing platform **100** may further include a proprietary database **334** that stores proprietary content **106A**. The example of FIG. **3** may be useful to identify user posts that contains proprietary source code, and to take actions based on configurable policies.

[0040] In an example, proprietary content **106A** includes proprietary source code of an entity/organization, external site **330** represents an external/public generative artificial intelligence machine learning (AIML) model, such as a large language model (LLM), and user post **110A** includes source code. In this example, a user may transmit source code to the generative AI model for training purposes and/or to have the generative AI model identify bugs, improve efficiency, and/or to provide alternative code. The example of FIG. **3** may be useful to identify source code of user post that contains proprietary source code, and to take actions based on configurable policies. The example of FIG. **3** is further described below with reference to FIG. **4**.

[0041] FIG. **4** illustrates a method **400** of protecting intellectual property through the use of digital signatures, according to an embodiment. Method **400** is described below with reference to FIG. **3**. Method **400** is not, however, limited to the example of FIG. **3**.

[0042] At **402**, signature generator **102** generates digital signatures **104** of proprietary content **106A**, such as described above with reference to **202** of method **200**.

[0043] At **404**, action tool **120** intercepts user post **110A** from user interface **114** (e.g., a browser). Action tool **120** may also pause the transmission of user post **110A** to external network **232** as a safety precaution.

[0044] At **406**, signature generator **102** generates a digital signature(s) **108A** of user post **110A**. Signature generator **102** may generate digital signature(s) **108A** as described above with respect to **204** of method **200**.

[0045] At **408**, signature matching engine **116** compares digital signature(s) **108A** of user post **110A** to digital signatures **104A** of proprietary content **106A** to identify matching digital signatures, such as described above with reference to **206** of method **200**. Thereafter, action tool **120** may perform one or more actions based on match results **118** and policies **128**, examples of which are provided below with reference to **410** through **424**.

[0046] At **410**, action tool **120** determines whether to initiate an audit of user post **110A**. Action tool **120** may determine to initiate an audit if match results **118** meet a criterion. As an example, match results **118** may provide a number/quantity of matching digital signatures, and the criterion may specify a threshold number of matching digital signatures. If the number/quantity of matching digital signatures meets the threshold, action tool **120** may determine to initiate an audit of user post **110A**. If action tool **120** determines to initiate an audit of user post **110A**, processing proceeds to **412**.

[0047] At **412**, action tool **120** may capture information for the audit. As an example, and without limitation, action tool **120** may capture user post **110A**, a user identifier associated with user post **110A**, a time-stamp associated with user post **110A**, digital signature **108A** of user post **110A**, matching digital signatures **104A**, and/or proprietary content **106A** corresponding to the matching digital signatures **104A**.

[0048] At **414**, action tool **120** may determine whether to send a notification to UI **114** based on match results **118** and policies **128**. As an example, action tool **120** may determine to send a notification if the number/quantity of matching digital signatures **104A** meet the threshold. If action tool **120** determines send a notification, processing proceeds to **416**.

[0049] At **416**, action tool **120** sends a notification to UI **114**. The notification may include a notice that user post **110A** contains or likely contains proprietary content (e.g., proprietary source code). The notification may also include a list of optional actions. If action tool **120** paused the transmission at **404**, the notification may include a link that permits the user to release (i.e., un-pause) or terminate the transmission.

[0050] At **418**, if action tool **120** paused the transmission of user post **110A** at **404**, action tool **120** may determine whether to release (i.e., un-pause) the transmission based on match results **118** and policies **128**. As an example, action tool **120** may determine to release the transmission if match results **118** do not meet the threshold. If action tool **120** determines to release the transmission, processing proceeds to **420**, where action tool **120** releases the transmission.

[0051] At **422**, if action tool **120** paused the transmission of user post **110A** at **404**, action tool **120** may determine whether to terminate the transmission based on match results **118** and policies **128**. As an example, action tool **120** may determine to terminate the transmission if match results **118** meet the threshold, or another threshold. If action tool **120** determines to terminate the transmission, processing proceeds to **424**, where action tool **120** terminates the transmission.

[0052] Action tool **120** may perform various combinations of actions described, a subset thereof, and/or other actions. Additional examples are provided below. Action tool **120** is not limited to the foregoing examples or the following examples.

[0053] In an example, action tool **120** pauses the transmission at **404**, releases the transmission if match results **118** do not meet a first threshold. Action tool **120** may further initiate an audit of user post **110A** and send an alert to UI **114**, such as described above. If match results **118** meet a second

threshold that is higher than the first threshold (i.e., indicating a higher likelihood that user post **110A** contains proprietary content), action tool **120** may terminate the transmission. If match results **118** fall between the first and second thresholds, action tool **120** may permit the user to select whether to release or terminate the transmission. In another example, action tool **120** does not pause the transmission of user post **110A** at **404**. Action tool **120** is not limited to the foregoing examples.

[0054] At **426**, processing may return to **404** for interception of other user posts and/or to **402** for generation of digital signatures **104A** of other proprietary content.

[0055] FIG. 5 is another block diagram of computing platform **100**, according to an embodiment. In the example of FIG. 5, reference content **106** includes ML posts **106B** of a machine-language (ML) model **532**, and user content **110** includes source code **110B** submitted to a source code repository **534**. In the example of FIG. 5, ML model **532** is illustrated as part of computing platform **100**. In another example, ML model **532** is hosted on another computing platform. Further in the example of FIG. 5, signature generator **102** may intercept source code **110B** from UI **114** and/or may retrieve source code **110B** from source code repository **534** (e.g., based on a routine/schedule and/or based on policies **124**).

[0056] In an example, ML model **532** includes a generative AI model (e.g., an LLM) that is trained to generate source code, and ML posts **106B** include ML-generated source code and/or responses to user queries regarding source code. The example of FIG. 5 may be useful to identify source code that contains ML posts, and to perform actions based on configurable policies. The example of FIG. 5 is described below with reference to FIG. 6.

[0057] FIG. 6 illustrates a method **600** of protecting intellectual property through the use of digital signatures, according to an embodiment. Method **600** is described below with reference to FIG. 5. Method **500** is not, however, limited to the example of FIG. 5.

[0058] At **602**, signature generator **102** generates digital signatures **104B** of ML posts **106B**, such as described above with reference to **202** of method **200**.

[0059] At **604**, signature generator **102** generates a digital signature(s) **108B** of source code **110B**, such as described above with respect to **204** of method **200**.

[0060] At **606**, signature matching engine **116** compares digital signature(s) **108B** of source code **110B** to digital signatures **104B** of ML posts **106B** to identify matching digital signatures, such as described above with reference to **206** of method **200**. Thereafter, action tool **120** may selectively perform one or more actions based on match results **118** and policies **128**, examples of which are provided below with reference to **608** through **614**.

[0061] At **608**, action tool **120** may determine whether to initiate an audit of source code **110B**, such as described above with reference to **410** of method **400**. If action tool **120** determines to initiate an audit of source code **110B**, processing proceeds to **610**, where action tool **120** may capture information for the audit, such as described above with reference to **412** of method **400**.

[0062] At **612**, action tool **120** may determine whether to send a notification to UI **114** based on match results **118** and policies **128**, such as described above with reference to **414** of method **400**. If action tool **120** determines to send a notification, processing proceeds to **614**.

[0063] At **614**, action tool **120** sends a notification to UI **114**. The notification may include a notice that source code **110B** contains or likely contains ML posts.

[0064] At **616**, processing may return to **604** for generation of digital signature(s) **108B** of other source code submitted to source code repository **534**, and/or to **602** for generation of digital signatures **104B** of other proprietary content.

[0065] Another example is a method that includes generating digital signatures of proprietary content, intercepting user content of a network transmission directed to an external network, generating a digital signature of the user content, by the computing platform, comparing the digital signature of the user content to the digital signatures of the proprietary content to identify matching digital signatures of the proprietary content, and selectively performing an action based on the

matching digital signatures and a configurable policy.

[0066] Another example is a non-transitory computer readable medium that includes stored instructions, which when executed by a processor, cause the processor to generate digital signatures of proprietary content, intercept user content of a network transmission directed to an external network, generate a digital signature of the user content, compare the digital signature of the user content to the digital signatures of the proprietary content to identify matching digital signatures of the proprietary content, and selectively perform an action based on the matching digital signatures and a configurable policy, where the action includes one or more of initiating an audit of the user content, sending a notification to a user interface indicating that the user content likely contains a portion of the proprietary content, releasing the transmission, and terminating the transmission.

[0067] Another example is a method that includes generating digital signatures of posts of a machine learning (ML) model, generating a digital signature of source code submitted to a source code repository, comparing the digital signature of the source code to the digital signatures of the posts of the ML model to identify matching digital signatures of the posts of the ML model, and selectively performing an action based on the matching digital signatures and a configurable policy, by the computing platform.

[0068] FIG. 7 is another block diagram of computing platform **100**, according to an embodiment. In the example of FIG. 7, reference content **106** includes proprietary content **106A** and ML posts **106B**, and user content **110** includes user post **110A** and source code **110B**. The example of FIG. 7 may be useful to identify user posts that contain proprietary source code and source code that contains posts of a ML model, and to take actions based on configurable policies.

[0069] FIG. 8 illustrates an example machine of a computer system **800** within which a set of instructions, for causing the machine to perform any one or more of the methodologies discussed herein, may be executed. In alternative implementations, the machine may be connected (e.g., networked) to other machines in a LAN, an intranet, an extranet, and/or the Internet. The machine may operate in the capacity of a server or a client machine in client-server network environment, as a peer machine in a peer-to-peer (or distributed) network environment, or as a server or a client machine in a cloud computing infrastructure or environment.

[0070] The machine may be a personal computer (PC), a tablet PC, a set-top box (STB), a Personal Digital Assistant (PDA), a cellular telephone, a web appliance, a server, a network router, a switch or bridge, or any machine capable of executing a set of instructions (sequential or otherwise) that specify actions to be taken by that machine. Further, while a single machine is illustrated, the term “machine” shall also be taken to include any collection of machines that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies discussed herein.

[0071] The example computer system **800** includes a processing device **802**, a main memory **804** (e.g., read-only memory (ROM), flash memory, dynamic random access memory (DRAM) such as synchronous DRAM (SDRAM), a static memory **806** (e.g., flash memory, static random access memory (SRAM), etc.), and a data storage device **818**, which communicate with each other via a bus **830**.

[0072] Processing device **802** represents one or more processors such as a microprocessor, a central processing unit, or the like. More particularly, the processing device may be complex instruction set computing (CISC) microprocessor, reduced instruction set computing (RISC) microprocessor, very long instruction word (VLIW) microprocessor, or a processor implementing other instruction sets, or processors implementing a combination of instruction sets. Processing device **802** may also be one or more special-purpose processing devices such as an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), a digital signal processor (DSP), network processor, or the like. The processing device **802** may be configured to execute instructions **826** for performing the operations and steps described herein.

[0073] The computer system **800** may further include a network interface device **808** to

communicate over the network **820**. The computer system **800** also may include a video display unit **810** (e.g., a liquid crystal display (LCD) or a cathode ray tube (CRT)), an alphanumeric input device **812** (e.g., a keyboard), a cursor control device **814** (e.g., a mouse), a graphics processing unit **822**, a signal generation device **816** (e.g., a speaker), graphics processing unit **822**, video processing unit **828**, and audio processing unit **832**.

[0074] The data storage device **818** may include a machine-readable storage medium **824** (also known as a non-transitory computer-readable medium) on which is stored one or more sets of instructions **826** or software embodying any one or more of the methodologies or functions described herein. The instructions **826** may also reside, completely or at least partially, within the main memory **804** and/or within the processing device **802** during execution thereof by the computer system **800**, the main memory **804** and the processing device **802** also constituting machine-readable storage media.

[0075] In some implementations, the instructions **826** include instructions to implement functionality corresponding to the present disclosure. While the machine-readable storage medium **824** is shown in an example implementation to be a single medium, the term “machine-readable storage medium” should be taken to include a single medium or multiple media (e.g., a centralized or distributed database, and/or associated caches and servers) that store the one or more sets of instructions. The term “machine-readable storage medium” shall also be taken to include any medium that is capable of storing or encoding a set of instructions for execution by the machine and that cause the machine and the processing device **802** to perform any one or more of the methodologies of the present disclosure. The term “machine-readable storage medium” shall accordingly be taken to include, but not be limited to, solid-state memories, optical media, and magnetic media.

[0076] Some portions of the preceding detailed descriptions have been presented in terms of algorithms and symbolic representations of operations on data bits within a computer memory. These algorithmic descriptions and representations are the ways used by those skilled in the data processing arts to most effectively convey the substance of their work to others skilled in the art. An algorithm may be a sequence of operations leading to a desired result. The operations are those requiring physical manipulations of physical quantities. Such quantities may take the form of electrical or magnetic signals capable of being stored, combined, compared, and otherwise manipulated. Such signals may be referred to as bits, values, elements, symbols, characters, terms, numbers, or the like.

[0077] It should be borne in mind, however, that all of these and similar terms are to be associated with the appropriate physical quantities and are merely convenient labels applied to these quantities. Unless specifically stated otherwise as apparent from the present disclosure, it is appreciated that throughout the description, certain terms refer to the action and processes of a computer system, or similar electronic computing device, that manipulates and transforms data represented as physical (electronic) quantities within the computer system's registers and memories into other data similarly represented as physical quantities within the computer system memories or registers or other such information storage devices.

[0078] The present disclosure also relates to an apparatus for performing the operations herein. This apparatus may be specially constructed for the intended purposes, or it may include a computer selectively activated or reconfigured by a computer program stored in the computer. Such a computer program may be stored in a computer readable storage medium, such as, but not limited to, any type of disk including floppy disks, optical disks, CD-ROMs, and magnetic-optical disks, read-only memories (ROMs), random access memories (RAMs), EPROMs, EEPROMs, magnetic or optical cards, or any type of media suitable for storing electronic instructions, each coupled to a computer system bus.

[0079] The algorithms and displays presented herein are not inherently related to any particular computer or other apparatus. Various other systems may be used with programs in accordance with

the teachings herein, or it may prove convenient to construct a more specialized apparatus to perform the method. In addition, the present disclosure is not described with reference to any particular programming language. It will be appreciated that a variety of programming languages may be used to implement the teachings of the disclosure as described herein.

[0080] The present disclosure may be provided as a computer program product, or software, that may include a machine-readable medium having stored thereon instructions, which may be used to program a computer system (or other electronic devices) to perform a process according to the present disclosure. A machine-readable medium includes any mechanism for storing information in a form readable by a machine (e.g., a computer). For example, a machine-readable (e.g., computer-readable) medium includes a machine (e.g., a computer) readable storage medium such as a read only memory (“ROM”), random access memory (“RAM”), magnetic disk storage media, optical storage media, flash memory devices, etc.

[0081] In the foregoing disclosure, implementations of the disclosure have been described with reference to specific example implementations thereof. It will be evident that various modifications may be made thereto without departing from the broader spirit and scope of implementations of the disclosure as set forth in the following claims. Where the disclosure refers to some elements in the singular tense, more than one element can be depicted in the figures and like elements are labeled with like numerals. The disclosure and drawings are, accordingly, to be regarded in an illustrative sense rather than a restrictive sense.

Claims

1. A method, comprising: generating reference digital signatures of first content; intercepting second content of a network transmission directed to an external network; generating a digital signature of the second content; comparing, by a processing device, the digital signature of the second content to the reference digital signatures of the first content to identify matching digital signatures of the first content; and selectively performing an action based on the matching digital signatures and a policy.
2. The method of claim 1, wherein: the second content comprises source code; and the external network comprises a generative artificial intelligence model (AIGL).
3. The method of claim 1, wherein the selectively performing an action comprises: initiating an audit of the second content if a number of the matching digital signatures meets a threshold of the policy.
4. The method of claim 3, wherein the initiating an audit comprises: capturing one or more of, the second content, a user identifier associated with the second content, a time-stamp associated with the second content, the digital signature of the second content, the matching digital signatures, and the first content corresponding to the matching digital signatures.
5. The method of claim 1, wherein the selectively performing an action comprises: sending a notification to a user interface indicating that the second content likely contains a portion of the first content if a number of the matching digital signatures meets a threshold of the policy.
6. The method of claim 1, further comprising: pausing the network transmission; generating the digital signature of the second content while the network transmission is paused; and comparing the digital signature of the second content to the reference digital signatures of the proprietary content while the network transmission is paused.
7. The method of claim 6, wherein the selectively performing an action comprises: releasing the transmission if a number of the matching digital signatures does not meet a threshold of the policy.
8. The method of claim 6, wherein the selectively performing an action comprises: terminating the transmission if a number of the matching digital signatures meets a threshold of the policy.
9. The method of claim 6, wherein the selectively performing an action comprises: sending a notification to a user interface if a number of the matching digital signatures meets a threshold of

the policy, wherein the notification comprises a notice that the second content likely contains a portion of the proprietary content and one or more of a link to permit a user to release the transmission and a link to permit the user to terminate the transmission.

10. The method of claim 6, wherein the selectively performing an action comprises: releasing the transmission if a number of the matching digital signatures does not meet a first threshold of the policy; initiating an audit of the second content if the number of the matching digital signatures meets the first threshold; and terminating the transmission if the number of the matching digital signatures meets a second threshold of the policy, wherein the second threshold is higher than the first threshold.

11. The method of claim 9, wherein the selectively performing an action further comprises: sending a notification to a user interface if the number of the matching digital signatures is between the first and second thresholds, wherein the notification comprises a notice that the second content likely contains a portion of the proprietary content, a link to permit a user to release the transmission, and a link to permit the user to terminate the transmission.

12. The method of claim 1, further comprising one or more of: generating the reference digital signatures and the digital signature of the second content based on a policy that specifies one or more of multiple methods of generating digital signatures; generating the reference digital signatures and the digital signature of the second content based on a policy that specifies one or more levels of granularity at which to generate the digital signatures; generating the reference digital signatures and the digital signature of the second content based on a content leakage risk identification policy; generating the reference digital signatures and the digital signature of the second content based on a policy that specifies content types for which digital signatures are to be generated; updating a database of the reference digital signatures based a policy that specifies one or more of an update frequency and an update schedule; and comparing the digital signature of the second content to the reference digital signatures of the proprietary content based on a policy that specifies one or more levels of digital signature granularity at which to compare the digital signatures.

13. The method of claim 1, wherein: the generating reference digital signatures comprises generating multiple reference digital signatures for a unit of the proprietary content at multiple respective levels of granularity based on features of the unit of the proprietary content; the generating a digital signature of the second content comprises generating multiple digital signatures for the second content at the multiple respective levels of granularity based on features of the second content; and the comparing comprises comparing one or more of the multiple reference digital signatures to one or more of the corresponding multiple digital signatures of the second content based on a policy; wherein the features of the unit of the proprietary content and the features of the second content comprise one or more of snippets of the respective content, metadata of the respective content, and directory structure descriptors of the respective content.

14. A non-transitory computer readable medium comprising stored instructions, which when executed by a processor, cause the processor to: generate reference digital signatures of posts of a machine learning (ML) model; generate a digital signature of source code submitted to a source code repository; compare the digital signature of the source code to the reference digital signatures of the posts of the ML model to identify matching digital signatures of the posts of the ML model; and selectively perform an action based on the matching digital signatures and a policy.

15. The non-transitory computer readable medium of claim 14, wherein the stored instructions, when executed, further cause the processor to selectively perform the action by: initiating an audit of the source code if a number of the matching digital signatures meets a threshold of the configurable policy.

16. The non-transitory computer readable medium of claim 15, wherein the stored instructions, when executed, further cause the processor to initiate the audit by: capturing one or more of, the source code, a user identifier associated with the source code, a time-stamp associated with the

source code, the digital signature of the source code, the matching digital signatures, and the ML posts corresponding to the matching digital signatures.

17. The non-transitory computer readable medium of claim 14, wherein the stored instructions, when executed, further cause the processor to selectively perform the action by: sending a notification to a user interface indicating that the source code likely contains a portion of the ML posts if a number of the matching digital signatures meets a threshold of the configurable policy.

18. The non-transitory computer readable medium of claim 14, wherein the stored instructions, when executed, further cause the processor to selectively perform the action by: initiating an audit of the source code and sending a notification to a user interface indicating that the source code likely contains a portion of the posts of the ML model, if a number of the matching digital signatures meets a threshold of the configurable policy.

19. A system comprising: a memory storing instructions; and a processor, coupled with the memory and to execute the instructions, the instructions when executed cause the processor to: generate reference digital signatures of first content; intercept second content of a network transmission directed to an external network; generate a digital signature of the second content; compare the digital signature of the second content to the reference digital signatures of the first content to identify matching digital signatures of the first content; selectively performing an action based on the matching digital signatures matching digital signatures of the first content and a policy; generate reference digital signatures of posts of a machine learning (ML) model; generate a digital signature of source code submitted to a source code repository; compare the digital signature of the source code to the reference digital signatures of the posts of the ML model to identify matching digital signatures of the posts of the ML model; and selectively perform an action based on the matching digital signatures of the posts of the ML and the policy.

20. The system of claim 19, wherein the action comprises one or more of: releasing the transmission if a number of the matching digital signatures of the first content does not meet a first threshold; initiating an audit of the user content if the number of the matching digital signatures of the first content meets the first threshold; sending a notification to a user interface indicating that the second content likely contains a portion of the first content if a number of the matching digital signatures of the first content meets the first threshold; terminating the transmission if the number of the matching digital signatures of the first content meets a second threshold; initiating an audit of the source code if a number of the matching digital signatures of the posts of the ML model meets a third threshold; and sending a notification to a user interface indicating that the source code likely contains a portion of the ML posts if the number of the matching digital signatures of the posts of the ML model meets the third threshold.
